Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mass” and “system” to explain your model or result. Sentence starters are optional.

mass: a measure of how strongly an object resists acceleration

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Bring back an earlier idea

A cart accelerates uniformly from rest to 12 m/s east in 4 s.
Find acceleration, average velocity and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = F_{net}/M = m_h\,g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass.
Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials.
Label what was held constant.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Read the sample pattern

Use the supplied ideal data. What happens to acceleration when mass doubles from 3 kg to 2 kg? Write a claim with two values.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

M (kg)	a (m/s²)
1	4
2	2
4	1

Marzano check - circle: ① ② ③

04 Plan a fair test

On a cart-and-hanger setup, name what you change, what you measure and two things you keep the same.
Explain why increasing the hanger mass too would spoil this comparison.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Record evidence honestly

If equipment is available, follow the teacher's setup and record three acceleration trials at each of two total masses. Otherwise use the supplied model data in 3 and label it model data.
Compare the results.

Support: My claim is ____. My two supporting values are ____.

Sentence starter: Both ... . They differ in ... because ... .

Total M	Trial 1	Trial 2	Trial 3
___	___	___	___
___	___	___	___

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg sign hangs still from one vertical cord.
Use g=10 N/kg.
Name both forces, their sources, magnitudes and directions.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

At the same net force, a 1 kg system accelerates at 6 m/s².
Predict acceleration for a 2 kg system.
Name the condition that makes the prediction valid.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2.
Predict acceleration for a 4 kg cart at the same net force. Then describe a fair trial that could challenge your prediction.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3